

Predictive_ProblemSet_2

Baibhab Chowdhury 0738

2026-02-05

Introduction

This assignment focuses on the application of linear regression techniques within a predictive analytics framework. The objective is to estimate model parameters, assess model assumptions, and interpret results using both numerical summaries and visual diagnostics. All analyses are conducted using R to ensure reproducibility.

```
rm(list = ls())
```

1. Problem to demonstrate that the population regression line is fixed, but least square regression line varies

Suppose the population regression line is given by $Y = 2 + 3x$, while the data comes from the model $y = 2 + 3x + \varepsilon$.

Step 1: For x in the range $[5, 10]$ graph the population regression line.

Step 2: Generate $x_i (i = 1, 2, \dots, n)$ from $\text{Uniform}(5, 10)$ and $\varepsilon_i (i = 1, 2, \dots, n)$ from $N(0, 4^2)$. Hence, compute $y_1, y_2, \dots, y_n$.

Step 3: On the basis of the data $(x_i, y_i) (i = 1, 2, \dots, n)$ generated in Step 2, report the least squares regression line.

Step 4: Repeat steps 2-3 five times. Graph the 5 least squares regression lines over the population regression line obtained in Step 1. Interpret the findings.

Take $n = 50$. Set the seed as $\text{seed} = 123$.

```
set.seed(123)
```

```
x1=seq(5,10,length.out=200);x1
```

```
## [1] 5.000000 5.025126 5.050251 5.075377 5.100503 5.125628  
5.150754
```

```
## [8] 5.175879 5.201005 5.226131 5.251256 5.276382 5.301508  
5.326633
```

```
## [15] 5.351759 5.376884 5.402010 5.427136 5.452261 5.477387  
5.502513
```

```
## [22] 5.527638 5.552764 5.577889 5.603015 5.628141 5.653266  
5.678392
```

```
## [29] 5.703518 5.728643 5.753769 5.778894 5.804020 5.829146  
5.854271
```

```
## [36] 5.879397 5.904523 5.929648 5.954774 5.979899 6.005025
```

6.030151						
## [43]	6.055276	6.080402	6.105528	6.130653	6.155779	6.180905
6.206030						
## [50]	6.231156	6.256281	6.281407	6.306533	6.331658	6.356784
6.381910						
## [57]	6.407035	6.432161	6.457286	6.482412	6.507538	6.532663
6.557789						
## [64]	6.582915	6.608040	6.633166	6.658291	6.683417	6.708543
6.733668						
## [71]	6.758794	6.783920	6.809045	6.834171	6.859296	6.884422
6.909548						
## [78]	6.934673	6.959799	6.984925	7.010050	7.035176	7.060302
7.085427						
## [85]	7.110553	7.135678	7.160804	7.185930	7.211055	7.236181
7.261307						
## [92]	7.286432	7.311558	7.336683	7.361809	7.386935	7.412060
7.437186						
## [99]	7.462312	7.487437	7.512563	7.537688	7.562814	7.587940
7.613065						
## [106]	7.638191	7.663317	7.688442	7.713568	7.738693	7.763819
7.788945						
## [113]	7.814070	7.839196	7.864322	7.889447	7.914573	7.939698
7.964824						
## [120]	7.989950	8.015075	8.040201	8.065327	8.090452	8.115578
8.140704						
## [127]	8.165829	8.190955	8.216080	8.241206	8.266332	8.291457
8.316583						
## [134]	8.341709	8.366834	8.391960	8.417085	8.442211	8.467337
8.492462						
## [141]	8.517588	8.542714	8.567839	8.592965	8.618090	8.643216
8.668342						
## [148]	8.693467	8.718593	8.743719	8.768844	8.793970	8.819095
8.844221						
## [155]	8.869347	8.894472	8.919598	8.944724	8.969849	8.994975
9.020101						
## [162]	9.045226	9.070352	9.095477	9.120603	9.145729	9.170854
9.195980						
## [169]	9.221106	9.246231	9.271357	9.296482	9.321608	9.346734
9.371859						
## [176]	9.396985	9.422111	9.447236	9.472362	9.497487	9.522613
9.547739						
## [183]	9.572864	9.597990	9.623116	9.648241	9.673367	9.698492
9.723618						
## [190]	9.748744	9.773869	9.798995	9.824121	9.849246	9.874372
9.899497						
## [197]	9.924623	9.949749	9.974874	10.000000		

y1=2+3*x1

plot(x1,y1,type='l',col="blue",lwd=3)

```

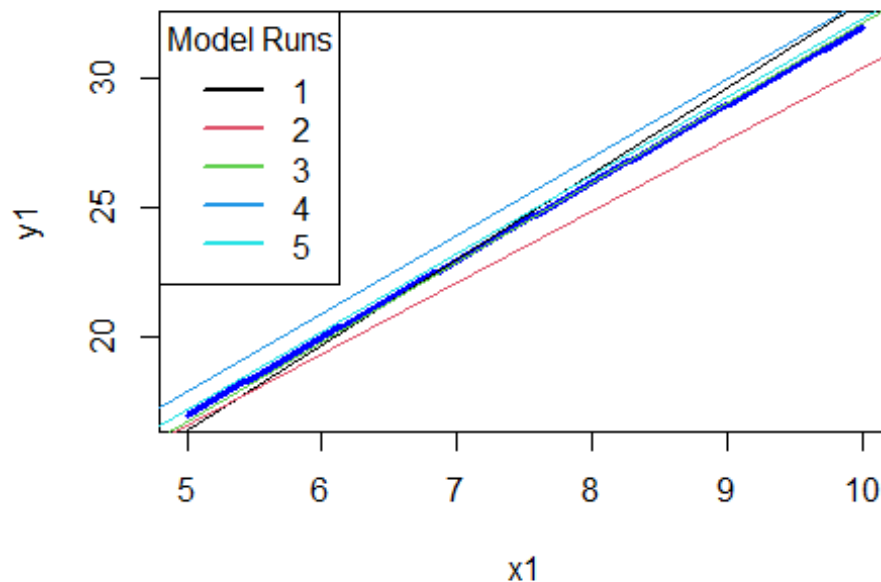
simulation_results <- data.frame(
  iteration = 1:5,
  intercept = numeric(5),
  slope = numeric(5)
)

for (i in 1:5){
  x = runif(50, 5, 10)
  eps = rnorm(50, 0, 4)
  y = 2 + 3*x + eps
  mod = lm(y ~ x)
  abline(mod, col = i)
  simulation_results$intercept[i] = coef(mod)[1]
  simulation_results$slope[i] = coef(mod)[2]
  print(summary(mod)$coefficients) # More detailed output
}

##               Estimate Std. Error    t value    Pr(>|t|)
## (Intercept) -0.09638929  2.8260956 -0.03410688 9.729334e-01
## x            3.30539569  0.3651861  9.05126412 5.961769e-12
##               Estimate Std. Error    t value    Pr(>|t|)
## (Intercept)  2.792188   3.7483426  0.7449128 4.599563e-01
## x            2.761042   0.4898233  5.6368122 8.950156e-07
##               Estimate Std. Error    t value    Pr(>|t|)
## (Intercept)  1.392997   2.7885513  0.4995416 6.196801e-01
## x            3.073267   0.3670719  8.3723847 6.062269e-11
##               Estimate Std. Error    t value    Pr(>|t|)
## (Intercept)  2.823089   2.7354376  1.032043  3.072243e-01
## x            3.023608   0.3623615  8.344175  6.681863e-11
##               Estimate Std. Error    t value    Pr(>|t|)
## (Intercept)  2.032506   3.1772492  0.6397063 5.254052e-01
## x            3.028097   0.4091446  7.4010440 1.789935e-09

legend("topleft",
      legend = paste(1:5),
      col = 1:5,
      lty = 1,
      lwd = 2,
      title = "Model Runs")

```



```
simulation_results
```

```
## iteration intercept slope
## 1 1 -0.09638929 3.305396
## 2 2 2.79218839 2.761042
## 3 3 1.39299737 3.073267
## 4 4 2.82308856 3.023608
## 5 5 2.03250638 3.028097
```

2. Problem to demonstrate that $\hat{\beta}_0$ and $\hat{\beta}$ minimizes RSS

Step 1: Generate x_i from Uniform(5, 10) and mean center the values. Generate ϵ_i from $N(0, 1)$. Calculate $y_i = 2 + 3x_i + \epsilon_i$, $i = 1, 2, \dots, n$. Take $n=50$ and seed=123.

Step 2: Now imagine that you only have the data on (x_i, y_i) , $i = 1, 2, \dots, n$, without knowing the mechanism that was used to generate the data in step 1. Assuming a linear regression of the type $y_i = \beta_0 + \beta x_i + \epsilon_i$, and based on these data (x_i, y_i) , $i = 1, 2, \dots, n$, obtain the least squares estimates of β_0 and β .

Step 3: Take a large number of grid values of (β_0, β) that also include the least squares estimates obtained from step 2. Compute the RSS for each parametric choice of (β_0, β) , where $RSS = (y_1 - \beta_0 - \beta x_1)^2 + (y_2 - \beta_0 - \beta x_2)^2 + \dots + (y_n - \beta_0 - \beta x_n)^2$. Find out for which combination of (β_0, β) , RSS is minimum.

```
n <- 50
xi <- runif(n, 5, 10)
xi_std <- xi - mean(xi)
```

```

eps <- rnorm(n, 0, 1)
y <- 2 + 3 * xi_std + eps

fitting <- lm(y ~ xi_std)
beta_not_hat <- coef(fitting)[1]
beta_hat <- coef(fitting)[2]

beta_not_grid <- seq(beta_not_hat - 2, beta_not_hat + 2, length = 51)
beta_grid <- seq(beta_hat - 2, beta_hat + 2, length = 51)

RSS <- matrix(NA, 50, 50)

for (i in 1:50) {
  for (j in 1:50) {
    RSS[i, j] <- sum((y - beta_not_grid[i] - beta_grid[j] * xi)^2)
  }
}
RSS

##           [,1]      [,2]      [,3]      [,4]      [,5]      [,6]      [,7]
## [1,] 2335.716 2684.182 3067.398 3485.366 3938.085 4425.556 4947.777
##      5504.750
## [2,] 2385.694 2738.797 3126.651 3549.256 4006.613 4498.721 5025.580
##      5587.190
## [3,] 2436.312 2794.052 3186.544 3613.786 4075.780 4572.526 5104.022
##      5670.270
## [4,] 2487.570 2849.948 3247.077 3678.957 4145.588 4646.970 5183.104
##      5753.989
## [5,] 2539.468 2906.483 3308.249 3744.767 4216.036 4722.055 5262.827
##      5838.349
## [6,] 2592.006 2963.659 3370.062 3811.217 4287.123 4797.780 5343.189
##      5923.349
## [7,] 2645.184 3021.474 3432.515 3878.307 4358.851 4874.145 5424.191
##      6008.989
## [8,] 2699.002 3079.929 3495.608 3946.037 4431.218 4951.150 5505.834
##      6095.268
## [9,] 2753.460 3139.025 3559.341 4014.408 4504.226 5028.795 5588.116
##      6182.188
## [10,] 2808.558 3198.760 3623.713 4083.418 4577.873 5107.080 5671.038
##      6269.748
## [11,] 2864.296 3259.136 3688.726 4153.068 4652.161 5186.005 5754.601
##      6357.948
## [12,] 2920.675 3320.151 3754.379 4223.358 4727.089 5265.570 5838.803
##      6446.787
## [13,] 2977.693 3381.807 3820.672 4294.288 4802.656 5345.775 5923.645
##      6536.267
## [14,] 3035.351 3444.102 3887.605 4365.859 4878.864 5426.620 6009.128
##      6626.387

```

[15,] 3093.649 3507.037 3955.177 4438.069 4955.711 5508.105 6095.250
6717.147
[16,] 3152.587 3570.613 4023.390 4510.919 5033.199 5590.230 6182.013
6808.546
[17,] 3212.165 3634.828 4092.243 4584.409 5111.326 5672.995 6269.415
6900.586
[18,] 3272.383 3699.684 4161.736 4658.539 5190.094 5756.400 6357.457
6993.266
[19,] 3333.241 3765.179 4231.869 4733.310 5269.502 5840.445 6446.140
7086.585
[20,] 3394.739 3831.314 4302.641 4808.720 5349.549 5925.130 6535.462
7180.545
[21,] 3456.877 3898.090 4374.054 4884.770 5430.237 6010.455 6625.424
7275.145
[22,] 3519.655 3965.505 4446.107 4961.460 5511.564 6096.420 6716.027
7370.385
[23,] 3583.073 4033.561 4518.800 5038.790 5593.532 6183.025 6807.269
7466.264
[24,] 3647.131 4102.256 4592.133 5116.760 5676.140 6270.270 6899.151
7562.784
[25,] 3711.829 4171.592 4666.105 5195.371 5759.387 6358.155 6991.674
7659.944
[26,] 3777.167 4241.567 4740.718 5274.621 5843.275 6446.680 7084.836
7757.744
[27,] 3843.145 4312.182 4815.971 5354.511 5927.802 6535.845 7178.638
7856.183
[28,] 3909.763 4383.438 4891.864 5435.041 6012.970 6625.650 7273.081
7955.263
[29,] 3977.021 4455.333 4968.397 5516.211 6098.777 6716.095 7368.163
8054.983
[30,] 4044.919 4527.869 5045.570 5598.022 6185.225 6807.180 7463.886
8155.343
[31,] 4113.457 4601.044 5123.382 5680.472 6272.313 6898.905 7560.248
8256.342
[32,] 4182.635 4674.860 5201.835 5763.562 6360.040 6991.270 7657.250
8357.982
[33,] 4252.453 4749.315 5280.928 5847.292 6448.408 7084.275 7754.893
8460.262
[34,] 4322.911 4824.410 5360.661 5931.662 6537.415 7177.920 7853.175
8563.182
[35,] 4394.009 4900.146 5441.034 6016.673 6627.063 7272.204 7952.097
8666.741
[36,] 4465.747 4976.521 5522.046 6102.323 6717.350 7367.129 8051.660
8770.941
[37,] 4538.125 5053.537 5603.699 6188.613 6808.278 7462.694 8151.862
8875.781
[38,] 4611.143 5131.192 5685.992 6275.543 6899.846 7558.899 8252.704
8981.261
[39,] 4684.801 5209.487 5768.925 6363.113 6992.053 7655.744 8354.187
9087.380

```

## [40,] 4759.099 5288.423 5852.498 6451.324 7084.901 7753.229 8456.309
9194.140
## [41,] 4834.037 5367.998 5936.710 6540.174 7178.388 7851.354 8559.071
9301.540
## [42,] 4909.615 5448.214 6021.563 6629.664 7272.516 7950.119 8662.474
9409.580
## [43,] 4985.833 5529.069 6107.056 6719.794 7367.284 8049.524 8766.516
9518.259
## [44,] 5062.692 5610.565 6193.189 6810.564 7462.691 8149.569 8871.198
9627.579
## [45,] 5140.190 5692.700 6279.962 6901.975 7558.739 8250.254 8976.521
9737.539
## [46,] 5218.328 5775.475 6367.374 6994.025 7655.426 8351.579 9082.483
9848.139
## [47,] 5297.106 5858.891 6455.427 7086.715 7752.754 8453.544 9189.086
9959.378
## [48,] 5376.524 5942.946 6544.120 7180.045 7850.721 8556.149 9296.328
10071.258
## [49,] 5456.582 6027.642 6633.453 7274.015 7949.329 8659.394 9404.210
10183.778
## [50,] 5537.280 6112.977 6723.426 7368.625 8048.577 8763.279 9512.733
10296.937
##           [,9]      [,10]      [,11]      [,12]      [,13]      [,14]      [,15]
## [1,] 6096.474 6722.950 7384.176 8080.154 8810.884 9576.364 10376.60
## [2,] 6183.551 6814.664 7480.528 8181.144 8916.510 9686.628 10491.50
## [3,] 6271.269 6907.019 7577.520 8282.773 9022.777 9797.532 10607.04
## [4,] 6359.626 7000.013 7675.152 8385.042 9129.684 9909.076 10723.22
## [5,] 6448.623 7093.648 7773.424 8487.952 9237.230 10021.260 10840.04
## [6,] 6538.260 7187.922 7872.336 8591.501 9345.417 10134.084 10957.50
## [7,] 6628.537 7282.837 7971.888 8695.690 9454.244 10247.548 11075.60
## [8,] 6719.454 7378.391 8072.080 8800.519 9563.710 10361.652 11194.35
## [9,] 6811.011 7474.586 8172.912 8905.989 9673.817 10476.397 11313.73
## [10,] 6903.208 7571.420 8274.384 9012.098 9784.564 10591.781 11433.75
## [11,] 6996.046 7668.895 8376.495 9118.847 9895.950 10707.805 11554.41
## [12,] 7089.523 7767.009 8479.247 9226.237 10007.977 10824.469 11675.71
## [13,] 7183.640 7865.764 8582.639 9334.266 10120.644 10941.773 11797.65
## [14,] 7278.397 7965.158 8686.671 9442.935 10233.950 11059.717 11920.23
## [15,] 7373.794 8065.193 8791.343 9552.244 10347.897 11178.301 12043.46
## [16,] 7469.831 8165.867 8896.655 9662.194 10462.484 11297.525 12167.32
## [17,] 7566.508 8267.182 9002.607 9772.783 10577.710 11417.389 12291.82
## [18,] 7663.826 8369.137 9109.199 9884.012 10693.577 11537.893 12416.96
## [19,] 7761.783 8471.731 9216.431 9995.882 10810.084 11659.037 12542.74
## [20,] 7860.380 8574.966 9324.303 10108.391 10927.230 11780.821 12669.16
## [21,] 7959.617 8678.840 9432.814 10221.540 11045.017 11903.245 12796.22
## [22,] 8059.494 8783.355 9541.966 10335.329 11163.444 12026.309 12923.93
## [23,] 8160.011 8888.509 9651.758 10449.759 11282.510 12150.013 13052.27
## [24,] 8261.168 8994.304 9762.190 10564.828 11402.217 12274.358 13181.25
## [25,] 8362.965 9100.738 9873.262 10680.537 11522.564 12399.342 13310.87
## [26,] 8465.403 9207.813 9984.974 10796.887 11643.551 12524.966 13441.13
## [27,] 8568.480 9315.527 10097.326 10913.876 11765.177 12651.230 13572.03

```

```

## [28,] 8672.197 9423.882 10210.318 11031.505 11887.444 12778.134 13703.57
## [29,] 8776.554 9532.876 10323.950 11149.775 12010.351 12905.678 13835.76
## [30,] 8881.551 9642.511 10438.222 11268.684 12133.897 13033.862 13968.58
## [31,] 8987.188 9752.785 10553.134 11388.233 12258.084 13162.686 14102.04
## [32,] 9093.465 9863.700 10668.685 11508.422 12382.911 13292.150 14236.14
## [33,] 9200.382 9975.254 10784.877 11629.252 12508.377 13422.254 14370.88
## [34,] 9307.940 10087.449 10901.709 11750.721 12634.484 13552.998 14506.26
## [35,] 9416.137 10200.283 11019.181 11872.830 12761.231 13684.382 14642.29
## [36,] 9524.974 10313.758 11137.293 11995.580 12888.617 13816.406 14778.95
## [37,] 9634.451 10427.872 11256.045 12118.969 13016.644 13949.070 14916.25
## [38,] 9744.568 10542.627 11375.437 12242.998 13145.311 14082.374 15054.19
## [39,] 9855.325 10658.021 11495.469 12367.667 13274.617 14216.319 15192.77
## [40,] 9966.722 10774.056 11616.141 12492.977 13404.564 14350.903 15331.99
## [41,] 10078.759 10890.730 11737.453 12618.926 13535.151 14486.127 15471.85
## [42,] 10191.437 11008.045 11859.404 12745.515 13666.377 14621.991 15612.36
## [43,] 10304.754 11125.999 11981.996 12872.745 13798.244 14758.495 15753.50
## [44,] 10418.711 11244.594 12105.228 13000.614 13930.751 14895.639 15895.28
## [45,] 10533.308 11363.828 12229.100 13129.123 14063.897 15033.423 16037.70
## [46,] 10648.545 11483.703 12353.612 13258.272 14197.684 15171.847 16180.76
## [47,] 10764.422 11604.217 12478.764 13388.062 14332.111 15310.911 16324.46
## [48,] 10880.939 11725.372 12604.556 13518.491 14467.177 15450.615 16468.80
## [49,] 10998.097 11847.167 12730.988 13649.560 14602.884 15590.959 16613.79
## [50,] 11115.894 11969.601 12858.060 13781.270 14739.231 15731.943 16759.41
##      [,16]  [,17]  [,18]  [,19]  [,20]  [,21]  [,22]
##      [,23]
## [1,] 11211.58 12081.31 12985.80 13925.04 14899.02 15907.76 16951.25
18029.49
## [2,] 11331.12 12205.49 13114.61 14058.49 15037.11 16050.49 17098.62
18181.50
## [3,] 11451.30 12330.31 13244.07 14192.58 15175.84 16193.85 17246.62
18334.14
## [4,] 11572.12 12455.76 13374.16 14327.31 15315.21 16337.86 17395.26
18487.42
## [5,] 11693.57 12581.86 13504.89 14462.68 15455.22 16482.51 17544.55
18641.34
## [6,] 11815.67 12708.59 13636.27 14598.69 15595.87 16627.79 17694.47
18795.90
## [7,] 11938.41 12835.97 13768.28 14735.34 15737.15 16773.72 17845.03
18951.10
## [8,] 12061.79 12963.99 13900.93 14872.63 15879.08 16920.28 17996.24
19106.94
## [9,] 12185.81 13092.64 14034.23 15010.56 16021.65 17067.49 18148.08
19263.42
## [10,] 12310.47 13221.94 14168.16 15149.13 16164.86 17215.33 18300.56
19420.54
## [11,] 12435.77 13351.88 14302.73 15288.35 16308.71 17363.82 18453.68
19578.30
## [12,] 12561.71 13482.45 14437.95 15428.20 16453.20 17512.95 18607.45
19736.70
## [13,] 12688.28 13613.67 14573.80 15568.69 16598.32 17662.71 18761.85

```


19895.74
[14,] 12815.50 13745.52 14710.30 15709.82 16744.09 17813.12 18916.89
20055.42
[15,] 12943.36 13878.02 14847.43 15851.59 16890.50 17964.16 19072.58
20215.74
[16,] 13071.86 14011.16 14985.20 15994.00 17037.55 18115.85 19228.90
20376.70
[17,] 13201.00 14144.93 15123.62 16137.05 17185.24 18268.17 19385.86
20538.30
[18,] 13330.78 14279.35 15262.67 16280.74 17333.57 18421.14 19543.47
20700.54
[19,] 13461.20 14414.41 15402.36 16425.07 17482.53 18574.75 19701.71
20863.42
[20,] 13592.26 14550.10 15542.70 16570.04 17632.14 18728.99 19860.59
21026.95
[21,] 13723.96 14686.44 15683.67 16715.66 17782.39 18883.88 20020.12
21191.11
[22,] 13856.29 14823.41 15825.28 16861.91 17933.28 19039.40 20180.28
21355.91
[23,] 13989.27 14961.03 15967.54 17008.80 18084.81 19195.57 20341.08
21521.35
[24,] 14122.89 15099.29 16110.43 17156.33 18236.98 19352.38 20502.53
21687.43
[25,] 14257.15 15238.18 16253.97 17304.50 18389.78 19509.82 20664.61
21854.15
[26,] 14392.05 15377.72 16398.14 17453.31 18543.23 19667.91 20827.33
22021.51
[27,] 14527.59 15517.89 16542.95 17602.76 18697.32 19826.63 20990.70
22189.51
[28,] 14663.77 15658.71 16688.41 17752.85 18852.05 19986.00 21154.70
22358.15
[29,] 14800.59 15800.17 16834.50 17903.58 19007.42 20146.00 21319.34
22527.43
[30,] 14938.05 15942.26 16981.23 18054.95 19163.43 20306.65 21484.62
22697.35
[31,] 15076.14 16085.00 17128.61 18206.97 19320.07 20467.94 21650.55
22867.91
[32,] 15214.88 16228.38 17276.62 18359.62 19477.36 20629.86 21817.11
23039.11
[33,] 15354.26 16372.39 17425.27 18512.91 19635.29 20792.43 21984.31
23210.95
[34,] 15494.28 16517.05 17574.57 18666.84 19793.86 20955.63 22152.16
23383.43
[35,] 15634.94 16662.34 17724.50 18821.41 19953.07 21119.48 22320.64
23556.55
[36,] 15776.24 16808.28 17875.07 18976.62 20112.92 21283.96 22489.76
23730.31
[37,] 15918.18 16954.86 18026.29 19132.47 20273.41 21449.09 22659.53
23904.71
[38,] 16060.76 17102.07 18178.14 19288.96 20434.53 21614.86 22829.93

24079.76
 ## [39,] 16203.97 17249.93 18330.64 19446.09 20596.30 21781.26 23000.97
 24255.44
 ## [40,] 16347.83 17398.43 18483.77 19603.86 20758.71 21948.31 23172.66
 24431.76
 ## [41,] 16492.33 17547.56 18637.54 19762.28 20921.76 22115.99 23344.98
 24608.72
 ## [42,] 16637.47 17697.34 18791.96 19921.33 21085.45 22284.32 23517.94
 24786.32
 ## [43,] 16783.25 17847.75 18947.01 20081.02 21249.78 22453.29 23691.55
 24964.56
 ## [44,] 16929.67 17998.81 19102.70 20241.35 21414.74 22622.89 23865.79
 25143.44
 ## [45,] 17076.73 18150.51 19259.04 20402.32 21580.35 22793.14 24040.67
 25322.96
 ## [46,] 17224.43 18302.84 19416.01 20563.93 21746.60 22964.02 24216.20
 25503.12
 ## [47,] 17372.77 18455.82 19573.62 20726.18 21913.49 23135.55 24392.36
 25683.92
 ## [48,] 17521.74 18609.44 19731.88 20889.07 22081.02 23307.71 24569.16
 25865.36
 ## [49,] 17671.36 18763.69 19890.77 21052.60 22249.19 23480.52 24746.60
 26047.44
 ## [50,] 17821.62 18918.59 20050.31 21216.77 22417.99 23653.97 24924.69
 26230.16
 ## [,24] [,25] [,26] [,27] [,28] [,29] [,30]
 [,31]
 ## [1,] 19142.49 20290.23 21472.73 22689.97 23941.97 25228.72 26550.22
 27906.47
 ## [2,] 19299.13 20451.51 21638.64 22860.52 24117.16 25408.55 26734.68
 28095.57
 ## [3,] 19456.40 20613.42 21805.19 23031.71 24292.99 25589.01 26919.79
 28285.31
 ## [4,] 19614.32 20775.98 21972.39 23203.54 24469.45 25770.12 27105.53
 28475.69
 ## [5,] 19772.88 20939.17 22140.22 23376.01 24646.56 25951.86 27291.91
 28666.71
 ## [6,] 19932.08 21103.01 22308.69 23549.12 24824.31 26134.25 27478.93
 28858.37
 ## [7,] 20091.92 21267.48 22477.80 23722.87 25002.70 26317.27 27666.60
 29050.67
 ## [8,] 20252.39 21432.60 22647.56 23897.27 25181.72 26500.94 27854.90
 29243.61
 ## [9,] 20413.51 21598.35 22817.95 24072.30 25361.39 26685.24 28043.84
 29437.19
 ## [10,] 20575.27 21764.75 22988.98 24247.97 25541.70 26870.19 28233.42
 29631.41
 ## [11,] 20737.67 21931.79 23160.65 24424.28 25722.65 27055.77 28423.64
 29826.27
 ## [12,] 20900.71 22099.46 23332.97 24601.23 25904.23 27242.00 28614.51

30021.77
[13,] 21064.38 22267.78 23505.92 24778.82 26086.46 27428.86 28806.01
30217.91
[14,] 21228.70 22436.73 23679.51 24957.05 26269.33 27616.37 28998.15
30414.69
[15,] 21393.66 22606.33 23853.75 25135.92 26452.84 27804.51 29190.93
30612.11
[16,] 21559.26 22776.56 24028.62 25315.43 26636.98 27993.29 29384.36
30810.17
[17,] 21725.49 22947.44 24204.13 25495.58 26821.77 28182.72 29578.42
31008.87
[18,] 21892.37 23118.95 24380.28 25676.37 27007.20 28372.78 29773.12
31208.21
[19,] 22059.89 23291.11 24557.08 25857.80 27193.27 28563.49 29968.46
31408.19
[20,] 22228.05 23463.90 24734.51 26039.87 27379.97 28754.83 30164.45
31608.81
[21,] 22396.85 23637.34 24912.58 26222.58 27567.32 28946.82 30361.07
31810.07
[22,] 22566.28 23811.41 25091.29 26405.93 27755.31 29139.44 30558.33
32011.97
[23,] 22736.36 23986.13 25270.65 26589.92 27943.94 29332.71 30756.23
32214.51
[24,] 22907.08 24161.48 25450.64 26774.55 28133.20 29526.61 30954.77
32417.69
[25,] 23078.44 24337.48 25631.27 26959.82 28323.11 29721.16 31153.96
32621.51
[26,] 23250.44 24514.12 25812.55 27145.73 28513.66 29916.34 31353.78
32825.97
[27,] 23423.07 24691.39 25994.46 27332.28 28704.85 30112.17 31554.24
33031.07
[28,] 23596.35 24869.31 26177.01 27519.47 28896.67 30308.63 31755.34
33236.80
[29,] 23770.27 25047.86 26360.20 27707.30 29089.14 30505.74 31957.09
33443.18
[30,] 23944.83 25227.06 26544.04 27895.77 29282.25 30703.48 32159.47
33650.20
[31,] 24120.03 25406.89 26728.51 28084.88 29476.00 30901.87 32362.49
33857.86
[32,] 24295.86 25587.37 26913.62 28274.63 29670.38 31100.89 32566.15
34066.16
[33,] 24472.34 25768.48 27099.37 28465.02 29865.41 31300.56 32770.45
34275.10
[34,] 24649.46 25950.24 27285.77 28656.05 30061.08 31500.86 32975.40
34484.68
[35,] 24827.22 26132.63 27472.80 28847.72 30257.39 31701.81 33180.98
34694.90
[36,] 25005.62 26315.67 27660.47 29040.03 30454.33 31903.39 33387.20
34905.76
[37,] 25184.65 26499.34 27848.79 29232.98 30651.92 32105.62 33594.06

```

35117.26
## [38,] 25364.33 26683.66 28037.74 29426.57 30850.15 32308.48 33801.57
35329.40
## [39,] 25544.65 26868.61 28227.33 29620.80 31049.02 32511.99 34009.71
35542.18
## [40,] 25725.61 27054.21 28417.56 29815.67 31248.52 32716.13 34218.49
35755.60
## [41,] 25907.21 27240.45 28608.44 30011.18 31448.67 32920.92 34427.91
35969.66
## [42,] 26089.44 27427.32 28799.95 30207.33 31649.46 33126.34 34637.98
36184.36
## [43,] 26272.32 27614.84 28992.10 30404.12 31850.89 33332.41 34848.68
36399.70
## [44,] 26455.84 27802.99 29184.89 30601.55 32052.95 33539.11 35060.02
36615.68
## [45,] 26640.00 27991.79 29378.33 30799.62 32255.66 33746.46 35272.00
36832.30
## [46,] 26824.79 28181.22 29572.40 30998.33 32459.01 33954.44 35484.62
37049.56
## [47,] 27010.23 28371.30 29767.11 31197.68 32663.00 34163.07 35697.89
37267.46
## [48,] 27196.31 28562.01 29962.47 31397.67 32867.62 34372.33 35911.79
37486.00
## [49,] 27383.03 28753.37 30158.46 31598.30 33072.89 34582.24 36126.33
37705.18
## [50,] 27570.39 28945.36 30355.09 31799.57 33278.80 34792.78 36341.51
37925.00
##      [,32]    [,33]    [,34]    [,35]    [,36]    [,37]    [,38]
[,39]
## [1,] 29297.48 30723.23 32183.74 33678.99 35209.00 36773.76 38373.27
40007.53
## [2,] 29491.21 30921.61 32386.75 33886.64 35421.29 36990.69 38594.83
40233.73
## [3,] 29685.59 31120.62 32590.40 34094.93 35634.22 37208.25 38817.03
40460.57
## [4,] 29880.61 31320.27 32794.69 34303.86 35847.78 37426.45 39039.88
40688.05
## [5,] 30076.27 31520.57 32999.62 34513.43 36061.99 37645.30 39263.36
40916.17
## [6,] 30272.56 31721.50 33205.20 34723.64 36276.84 37864.78 39487.48
41144.93
## [7,] 30469.50 31923.08 33411.41 34934.49 36492.32 38084.91 39712.24
41374.33
## [8,] 30667.08 32125.29 33618.26 35145.98 36708.45 38305.67 39937.64
41604.37
## [9,] 30865.29 32328.15 33825.75 35358.11 36925.21 38527.07 40163.68
41835.04
## [10,] 31064.15 32531.64 34033.88 35570.88 37142.62 38749.12 40390.36
42066.36
## [11,] 31263.65 32735.78 34242.66 35784.29 37360.67 38971.80 40617.69

```

42298.32
[12,] 31463.78 32940.55 34452.07 35998.34 37579.35 39195.13 40845.65
42530.92
[13,] 31664.56 33145.96 34662.12 36213.02 37798.68 39419.09 41074.25
42764.16
[14,] 31865.98 33352.02 34872.81 36428.35 38018.65 39643.69 41303.49
42998.04
[15,] 32068.04 33558.71 35084.14 36644.32 38239.25 39868.94 41533.37
43232.56
[16,] 32270.73 33766.05 35296.11 36860.93 38460.50 40094.82 41763.89
43467.72
[17,] 32474.07 33974.02 35508.73 37078.18 38682.39 40321.35 41995.05
43703.51
[18,] 32678.05 34182.64 35721.98 37296.07 38904.91 40548.51 42226.86
43939.95
[19,] 32882.66 34391.89 35935.87 37514.60 39128.08 40776.31 42459.30
44177.03
[20,] 33087.92 34601.79 36150.40 37733.77 39351.89 41004.76 42692.38
44414.75
[21,] 33293.82 34812.32 36365.57 37953.58 39576.33 41233.84 42926.10
44653.11
[22,] 33500.35 35023.49 36581.39 38174.03 39801.42 41463.56 43160.46
44892.11
[23,] 33707.53 35235.31 36797.84 38395.12 40027.15 41693.93 43395.46
45131.75
[24,] 33915.35 35447.76 37014.93 38616.85 40253.51 41924.93 43631.10
45372.03
[25,] 34123.81 35660.86 37232.66 38839.21 40480.52 42156.58 43867.38
45612.94
[26,] 34332.90 35874.59 37451.03 39062.22 40708.17 42388.86 44104.31
45854.50
[27,] 34542.64 36088.97 37670.04 39285.87 40936.45 42621.78 44341.87
46096.70
[28,] 34753.02 36303.98 37889.70 39510.16 41165.38 42855.35 44580.07
46339.54
[29,] 34964.03 36519.64 38109.99 39735.09 41394.95 43089.55 44818.91
46583.02
[30,] 35175.69 36735.93 38330.92 39960.66 41625.15 43324.40 45058.39
46827.14
[31,] 35387.99 36952.86 38552.49 40186.87 41856.00 43559.88 45298.51
47071.90
[32,] 35600.93 37170.44 38774.70 40413.72 42087.49 43796.00 45539.27
47317.30
[33,] 35814.50 37388.65 38997.56 40641.21 42319.61 44032.77 45780.68
47563.33
[34,] 36028.72 37607.51 39221.05 40869.34 42552.38 44270.17 46022.72
47810.01
[35,] 36243.58 37827.00 39445.18 41098.11 42785.79 44508.22 46265.40
48057.33
[36,] 36459.07 38047.14 39669.95 41327.52 43019.83 44746.90 46508.72

```

48305.29
## [37,] 36675.21 38267.91 39895.36 41557.57 43254.52 44986.22 46752.68
48553.89
## [38,] 36891.99 38489.33 40121.41 41788.25 43489.85 45226.19 46997.28
48803.13
## [39,] 37109.40 38711.38 40348.11 42019.58 43725.81 45466.79 47242.52
49053.01
## [40,] 37327.46 38934.07 40575.44 42251.55 43962.42 45708.04 47488.41
49303.53
## [41,] 37546.16 39157.41 40803.41 42484.16 44199.67 45949.92 47734.93
49554.68
## [42,] 37765.50 39381.38 41032.02 42717.41 44437.55 46192.44 47982.09
49806.48
## [43,] 37985.47 39606.00 41261.27 42951.30 44676.08 46435.61 48229.89
50058.92
## [44,] 38206.09 39831.25 41491.17 43185.83 44915.25 46679.41 48478.33
50312.00
## [45,] 38427.35 40057.15 41721.70 43421.00 45155.05 46923.86 48727.41
50565.72
## [46,] 38649.24 40283.68 41952.87 43656.81 45395.50 47168.94 48977.13
50820.08
## [47,] 38871.78 40510.86 42184.68 43893.26 45636.59 47414.66 49227.49
51075.08
## [48,] 39094.96 40738.67 42417.13 44130.35 45878.31 47661.03 49478.50
51330.72
## [49,] 39318.78 40967.12 42650.22 44368.08 46120.68 47908.03 49730.14
51586.99
## [50,] 39543.23 41196.22 42883.96 44606.44 46363.68 48155.68 49982.42
51843.91
##      [,40]    [,41]    [,42]    [,43]    [,44]    [,45]    [,46]
[,47]
## [1,] 41676.55 43380.31 45118.83 46892.09 48700.11 50542.88 52420.40
54332.67
## [2,] 41907.38 43615.79 45358.94 47136.84 48949.50 50796.90 52679.06
54595.97
## [3,] 42138.86 43851.90 45599.69 47382.23 49199.52 51051.57 52938.36
54859.91
## [4,] 42370.98 44088.65 45841.08 47628.26 49450.19 51306.87 53198.30
55124.49
## [5,] 42603.73 44326.05 46083.11 47874.93 49701.50 51562.81 53458.88
55389.71
## [6,] 42837.13 44564.08 46325.78 48122.24 49953.44 51819.40 53720.10
55655.56
## [7,] 43071.16 44802.75 46569.09 48370.18 50206.03 52076.62 53981.97
55922.06
## [8,] 43305.84 45042.07 46813.04 48618.77 50459.25 52334.48 54244.47
56189.20
## [9,] 43541.16 45282.02 47057.63 48868.00 50713.12 52592.99 54507.61
56456.98
## [10,] 43777.11 45522.61 47302.87 49117.87 50967.62 52852.13 54771.39

```

56725.40
[11,] 44013.71 45763.85 47548.74 49368.38 51222.77 53111.91 55035.81
56994.45
[12,] 44250.94 46005.72 47795.25 49619.53 51478.55 53372.34 55300.87
57264.15
[13,] 44488.82 46248.23 48042.40 49871.31 51734.98 53633.40 55566.57
57534.49
[14,] 44727.34 46491.39 48290.19 50123.74 51992.05 53895.10 55832.91
57805.47
[15,] 44966.49 46735.18 48538.62 50376.81 52249.75 54157.44 56099.89
58077.08
[16,] 45206.29 46979.61 48787.69 50630.52 52508.10 54420.43 56367.51
58349.34
[17,] 45446.73 47224.69 49037.40 50884.87 52767.08 54684.05 56635.77
58622.24
[18,] 45687.80 47470.40 49287.75 51139.86 53026.71 54948.31 56904.67
58895.78
[19,] 45929.52 47716.76 49538.74 51395.48 53286.97 55213.22 57174.21
59169.96
[20,] 46171.87 47963.75 49790.37 51651.75 53547.88 55478.76 57444.39
59444.77
[21,] 46414.87 48211.38 50042.65 51908.66 53809.43 55744.94 57715.21
59720.23
[22,] 46658.51 48459.66 50295.56 52166.21 54071.61 56011.77 57986.67
59996.33
[23,] 46902.78 48708.57 50549.11 52424.40 54334.44 56279.23 58258.77
60273.07
[24,] 47147.70 48958.12 50803.30 52683.23 54597.90 56547.33 58531.51
60550.44
[25,] 47393.25 49208.32 51058.13 52942.69 54862.01 56816.08 58804.89
60828.46
[26,] 47639.45 49459.15 51313.60 53202.80 55126.75 57085.46 59078.91
61107.12
[27,] 47886.29 49710.62 51569.71 53463.55 55392.14 57355.48 59353.57
61386.42
[28,] 48133.76 49962.74 51826.46 53724.94 55658.17 57626.14 59628.87
61666.36
[29,] 48381.88 50215.49 52083.85 53986.97 55924.83 57897.45 59904.82
61946.93
[30,] 48630.64 50468.88 52341.88 54249.63 56192.14 58169.39 60181.40
62228.15
[31,] 48880.03 50722.92 52600.55 54512.94 56460.08 58441.97 60458.62
62510.01
[32,] 49130.07 50977.59 52859.87 54776.89 56728.67 58715.20 60736.48
62792.51
[33,] 49380.74 51232.90 53119.82 55041.48 56997.89 58989.06 61014.98
63075.65
[34,] 49632.06 51488.86 53380.41 55306.71 57267.76 59263.56 61294.12
63359.42
[35,] 49884.02 51745.45 53641.64 55572.58 57538.27 59538.71 61573.90

```

63643.84
## [36,] 50136.61 52002.68 53903.51 55839.08 57809.41 59814.49 61854.32
63928.90
## [37,] 50389.85 52260.56 54166.02 56106.23 58081.20 60090.91 62135.38
64214.60
## [38,] 50643.72 52519.07 54429.17 56374.02 58353.62 60367.98 62417.08
64500.93
## [39,] 50898.24 52778.23 54692.96 56642.45 58626.69 60645.68 62699.42
64787.91
## [40,] 51153.40 53038.02 54957.39 56911.52 58900.39 60924.02 62982.40
65075.53
## [41,] 51409.19 53298.45 55222.46 57181.23 59174.74 61203.00 63266.02
65363.79
## [42,] 51665.63 53559.53 55488.17 57451.57 59449.73 61482.63 63550.28
65652.69
## [43,] 51922.70 53821.24 55754.53 57722.56 59725.35 61762.89 63835.18
65942.22
## [44,] 52180.42 54083.59 56021.52 57994.19 60001.62 62043.79 64120.72
66232.40
## [45,] 52438.78 54346.59 56289.15 58266.46 60278.52 62325.34 64406.90
66523.22
## [46,] 52697.77 54610.22 56557.42 58539.37 60556.07 62607.52 64693.72
66814.68
## [47,] 52957.41 54874.49 56826.33 58812.92 60834.25 62890.34 64981.18
67106.78
## [48,] 53217.69 55139.41 57095.88 59087.10 61113.08 63173.81 65269.28
67399.51
## [49,] 53478.60 55404.96 57366.07 59361.93 61392.55 63457.91 65558.02
67692.89
## [50,] 53740.16 55671.15 57636.90 59637.40 61672.65 63742.65 65847.41
67986.91
##      [,48]      [,49]      [,50]
## [1,] 56279.70 58261.47 60278.00
## [2,] 56547.63 58534.04 60555.21
## [3,] 56816.21 58807.26 60833.06
## [4,] 57085.42 59081.11 61111.55
## [5,] 57355.28 59355.60 61390.68
## [6,] 57625.77 59630.74 61670.45
## [7,] 57896.91 59906.51 61950.86
## [8,] 58168.68 60182.92 62231.91
## [9,] 58441.10 60459.97 62513.60
## [10,] 58714.15 60737.67 62795.93
## [11,] 58987.85 61016.00 63078.90
## [12,] 59262.19 61294.97 63362.51
## [13,] 59537.16 61574.58 63646.76
## [14,] 59812.78 61854.84 63931.65
## [15,] 60089.03 62135.73 64217.18
## [16,] 60365.93 62417.26 64503.35
## [17,] 60643.46 62699.43 64790.16
## [18,] 60921.64 62982.25 65077.61

```



```
## [19,] 61200.45 63265.70 65365.70
## [20,] 61479.91 63549.79 65654.43
## [21,] 61760.00 63834.52 65943.80
## [22,] 62040.74 64119.90 66233.81
## [23,] 62322.11 64405.91 66524.46
## [24,] 62604.13 64692.56 66815.75
## [25,] 62886.78 64979.86 67107.68
## [26,] 63170.08 65267.79 67400.25
## [27,] 63454.01 65556.36 67693.46
## [28,] 63738.59 65845.57 67987.31
## [29,] 64023.80 66135.43 68281.80
## [30,] 64309.66 66425.92 68576.93
## [31,] 64596.15 66717.05 68872.70
## [32,] 64883.29 67008.82 69169.11
## [33,] 65171.07 67301.24 69466.16
## [34,] 65459.48 67594.29 69763.85
## [35,] 65748.54 67887.98 70062.18
## [36,] 66038.23 68182.31 70361.15
## [37,] 66328.57 68477.29 70660.76
## [38,] 66619.54 68772.90 70961.01
## [39,] 66911.16 69069.15 71261.90
## [40,] 67203.41 69366.04 71563.43
## [41,] 67496.31 69663.58 71865.60
## [42,] 67789.84 69961.75 72168.41
## [43,] 68084.02 70260.56 72471.86
## [44,] 68378.83 70560.01 72775.95
## [45,] 68674.29 70860.11 73080.68
## [46,] 68970.38 71160.84 73386.05
## [47,] 69267.12 71462.21 73692.06
## [48,] 69564.49 71764.23 73998.71
## [49,] 69862.51 72066.88 74306.00
## [50,] 70161.16 72370.17 74613.93

which(RSS == min(RSS), arr.ind = TRUE)

##      row col
## [1,]    1   1
```

3. Problem to demonstrate that least square estimators are unbiased

Step 1: Generate $x_i (i = 1, 2, \dots, n)$ from $\text{Uniform}(0, 1)$, $\varepsilon_i (i = 1, 2, \dots, n)$ from $N(0, 1)$ and hence generate y using $y_i = \beta_0 + \beta x_i + \varepsilon_i$. (Take $\beta_0 = 2$, $\beta = 3$).

Step 2: On the basis of the data $(x_i, y_i) (i = 1, 2, \dots, n)$ generated in Step 1, obtain the least square estimates of β_0 and β .

Repeat Steps 1-2, $R = 1000$ times. In each simulation obtain $\hat{\beta}_0$ and $\hat{\beta}$. Finally, the least-square estimates will be given by the average of these estimated values. Compare these with the true β_0 and β and comment.

Take $n = 50$ and $\text{seed} = 123$.

```
set.seed(123)
n3 <- 50
R <- 10000

beta_not_hat_vec <- array()
beta_hat_vec <- array()
eps3 <- rnorm(n3,0,1)

for(r in 1:R){
  x3 <- runif(n3,0,1)
  y3 <- 2 + 3*x3 + eps3

  fit3 <- lm(y3 ~ x3)
  beta_not_hat_vec[r] <- coef(fit3)[1]
  beta_hat_vec[r] <- coef(fit3)[2]
}

predicted_beta_not_hat <- sum(beta_not_hat_vec)/R
predicted_beta_not_hat

## [1] 2.033112

predicted_beta_hat_vec <- sum(beta_hat_vec)/R
predicted_beta_hat_vec

## [1] 3.00193
```

4. Comparing several simple linear regressions

Attach “Boston” data from MASS library in R. Select median value of owner-occupied homes, as the response and per capita crime rate, nitrogen oxides concentration, proportion of blacks and percentage of lower status of the population as predictors.

- Selecting the predictors one by one, run four separate linear regressions to the data. Present the output in a single table.
- Which model gives the best fit?
- Compare the coefficients of the predictors from each model and comment on the usefulness of the predictors.

```
library(MASS)
data(Boston)

fit1 <- lm(medv ~ crim, data=Boston)
fit2 <- lm(medv ~ nox, data=Boston)
fit3 <- lm(medv ~ black, data=Boston)
fit4 <- lm(medv ~ lstat, data=Boston)

summary_t <- data.frame(
```

```

Model = c("crim", "nox", "black", "lstat"),
Intercept = c(coef(fit1)[1], coef(fit2)[1], coef(fit3)[1], coef(fit4)[1]),
Slope = c(coef(fit1)[2], coef(fit2)[2], coef(fit3)[2], coef(fit4)[2]),
R2 = c(summary(fit1)$r.squared,
        summary(fit2)$r.squared,
        summary(fit3)$r.squared,
        summary(fit4)$r.squared)
)

print(summary_t)

##      Model Intercept      Slope      R2
## crim    crim  24.03311 -0.41519028 0.1507805
## nox      nox  41.34587 -33.91605501 0.1826030
## black black  10.55103  0.03359306 0.1111961
## lstat lstat  34.55384 -0.95004935 0.5441463

best_model <- summary_t[which.max(summary_t$R2), ]
best_model

##      Model Intercept      Slope      R2
## lstat lstat  34.55384 -0.9500494 0.5441463

```